

Dynamic Multi-Agent RAG Workflow:

Execution Trace

This document provides a step-by-step execution trace of a complex, multi-part user query flowing through the re-engineered Multi-User Multi-Domain Agentic RAG System. The architecture now features **Dynamic Re-planning (ReAct)**, allowing the Query-Planning Agent to iteratively generate tasks, evaluate summaries, and dynamically route the workflow until the query is fully resolved.

Step 0: Initialization (Query Entry)

Event: The system initializes the starting state. The query requires both a semantic vector search (policy rules) and a structured database lookup (user balance). The state is passed to the Supervisor.

```
{
  "messages": [
    {
      "role": "user",
      "content": "What is the HR leave policy for taking a week off, and do I have enough leave balance to do so?"
    }
  ],
  "next_agent": null,
  "step_count": 0,
  "allowed_domains": ["HR", "IT"],
  "query_intent": null,
  "tasks": [],
  "current_task_id": null,
  "retrieved_context": [],
  "is_context_valid": null,
  "verification_feedback": null,
  "citations": null
}
```

Step 1: Query-Planning Agent (Initial Classification & First Task)

Event: The Supervisor sees an empty `query_intent` and routes to the Planner. The Planner classifies the intent and generates only the very first task to avoid hallucinating dependencies.

```
{
  ... // existing state
  "query_intent": "information_retrieval",
  "tasks": [
    {
      "task_id": "task_1",
      "description": "Search HR policy documents to determine the rules and requirements for taking a continuous one-week leave.",
      "task_type": "vector_search",
      "target_domain": "HR",
      "status": "pending",
      "result_summary": null
    }
  ],
  "current_task_id": "task_1",
  "next_agent": "Supervisor",
  "step_count": 1
}
```

Step 2: Supervisor Delegates Task 1 (Vector Search)

Event: The Supervisor inspects `current_task_id`, identifies it as a `vector_search`, and routes the state to the Retrieval Agent.

```
{
  ... // existing state
  "next_agent": "Retrieval_Agent",
  "step_count": 2
}
```

Step 3: Retrieval Agent & Verification (HR Data)

Event: The Retrieval Agent queries Qdrant for HR policies, appends the chunks, and updates the task status. The Verification Agent approves it.

```
{
  ... // existing state
  "tasks": [
    {
      "task_id": "task_1",
      "task_type": "vector_search",
      "status": "completed",
      "result_summary": "HR policy allows up to 2 weeks of continuous leave provided the employee has sufficient leave balance and manager approval."
    }
  ],
  "current_task_id": null,
  "retrieved_context": [
    {
      "chunk_id": "chk_hr_leave_04",
      "file_name": "HR_Leave_Policy_2026.pdf",
      "text": "Employees may request continuous annual leave of up to 14 days (2 weeks) subject to available balance.",
      "page": 4
    }
  ],
  "is_context_valid": true,
  "next_agent": "Supervisor",
  "step_count": 4
}
```

Step 4: Query-Planning Agent (Dynamic Re-planning)

Event: With `current_task_id` set to null, the Supervisor loops back to the Planner. The Planner reads the `result_summary` of task 1, remembers the original query about the user's specific balance, and generates task 2.

```
{
  ... // existing state
```

```

"tasks": [
  {
    "task_id": "task_1",
    "status": "completed",
    "result_summary": "HR policy allows up to 2 weeks of continuous leave provided the
employee has sufficient leave balance and manager approval."
  },
  {
    "task_id": "task_2",
    "description": "Perform a structured DB lookup to retrieve the current available leave
balance for the user.",
    "task_type": "structured_db_lookup",
    "target_domain": "HR",
    "status": "pending",
    "result_summary": null
  }
],
"current_task_id": "task_2",
"next_agent": "Supervisor",
"step_count": 5
}

```

Step 5: Structured-Data Agent & Verification (DB Lookup)

Event: The Supervisor routes to the Structured-Data Agent. The agent queries the SQL DB, fetches the balance, and updates the task summary.

```

{
  ... // existing state
  "tasks": [
    {
      "task_id": "task_1",
      "status": "completed"
    },
    {
      "task_id": "task_2",
      "task_type": "structured_db_lookup",
      "status": "completed",

```

```

    "result_summary": "The user currently has 12 days of available leave balance."
  }
],
"current_task_id": null,
"retrieved_context": [
  {
    "chunk_id": "chk_hr_leave_04",
    "file_name": "HR_Leave_Policy_2026.pdf",
    "text": "Employees may request continuous annual leave of up to 14 days (2 weeks)
subject to available balance.",
    "page": 4
  },
  {
    "chunk_id": "db_record_leave_bal",
    "file_name": "HR_SQL_Database",
    "text": "User: Active | Leave Balance: 12 Days",
    "page": null
  }
],
"is_context_valid": true,
"next_agent": "Supervisor",
"step_count": 7
}

```

Step 6: Query-Planning Agent (Final Evaluation)

Event: The Planner evaluates the state, sees that both the policy conditions and the user's balance have been retrieved, and concludes that no further tasks are needed.

```

{
  ... // existing state
  "current_task_id": "READY_FOR_SYNTHESIS",
  "next_agent": "Supervisor",
  "step_count": 8
}

```

Step 7: Synthesis Agent Drafts Final Answer (Final

Output)

Event: The Supervisor reads the ready flag and routes to the Synthesis Agent. The agent formulates the answer, maps the citations, and concludes the workflow.

```
{
  "messages": [
    {
      "role": "user",
      "content": "What is the HR leave policy for taking a week off, and do I have enough leave balance to do so?"
    },
    {
      "role": "assistant",
      "content": "According to the HR Leave Policy, you are permitted to take up to 2 weeks (14 days) of continuous leave, provided you have sufficient balance and manager approval. I have checked your current records, and you have 12 days of leave balance available, which is perfectly sufficient to take a week off."
    }
  ],
  "next_agent": "END",
  "step_count": 9,
  "citations": [
    {
      "span": "permitted to take up to 2 weeks (14 days) of continuous leave",
      "source_file": "HR_Leave_Policy_2026.pdf",
      "chunk_id": "chk_hr_leave_04",
      "page": 4
    },
    {
      "span": "you have 12 days of leave balance available",
      "source_file": "HR_SQL_Database",
      "chunk_id": "db_record_leave_bal",
      "page": null
    }
  ]
}
```